

WATERTOWN, WI, USA

WMO: 726464

Lat: 43.167N

Lon: 88.717W

Elev: 250

StdP: 98.36

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 54834

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.3	-18.6	-26.2	0.4	-20.4	-23.0	0.5	-17.4	10.1	-5.3	9.0	-3.9	2.1	280	0.520

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	32.1	24.0	30.2	23.2	28.7	22.5	26.2	29.4	24.9	28.3	23.8	27.2	4.4	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.0	28.0	23.8	19.3	26.8	22.7	17.9	25.7	83.0	29.5	76.8	28.1	72.6	27.5	31.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.2	7.3	DB	-25.6	34.1	3.5	1.9	-28.1	35.5	-30.2	36.6	-32.1	37.7	-34.7	39.1	(4)
(5)			WB	-25.9	28.1	3.4	1.5	-28.3	29.2	-30.3	30.0	-32.3	30.9	-34.7	32.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	-6.6	-4.8	1.3	8.0	14.3	19.7	22.0	21.0	17.0	10.2	2.9	-3.6	(6)	
	DBStd	11.29	6.60	6.70	6.47	5.25	4.59	3.74	3.17	2.98	4.39	5.08	5.56	6.04	(7)	
	HDD10.0	2028	515	414	282	97	12	0	0	0	3	61	222	423	(8)	
	HDD18.3	3921	773	647	530	311	142	27	5	9	75	257	464	681	(9)	
	CDD10.0	1487	0	1	11	38	145	292	372	339	214	67	7	1	(10)	
	CDD18.3	341	0	0	1	2	17	69	119	91	36	5	0	0	(11)	
	CDH23.3	3143	0	0	7	39	192	666	1123	727	337	52	1	0	(12)	
	CDH26.7	910	0	0	1	7	46	203	368	189	88	8	0	0	(13)	
(14) Wind	WSAvg	3.0	3.3	3.3	3.4	3.8	3.3	2.6	2.3	2.0	2.4	3.0	3.2	3.2	(14)	
(15) Precipitation	PrecAvg	850	33	29	51	76	85	108	106	102	97	63	58	40	(15)	
	PrecMax	1120	88	85	128	177	246	238	266	233	343	160	155	98	(16)	
	PrecMin	499	2	0	6	21	19	30	18	14	9	6	5	6	(17)	
	PrecStd	130	20	21	31	32	48	57	50	50	69	39	32	24	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	13.9	23.8	27.6	30.9	32.9	35.2	32.6	31.7	27.8	20.3	13.7	(19)	
		MCWB	8.6	10.6	17.6	16.8	21.3	23.8	25.7	25.4	22.7	19.8	14.8	12.9	(20)	
	2%	DB	6.1	8.9	17.9	23.7	27.7	31.0	32.2	30.7	29.0	24.3	16.7	8.8	(21)	
		MCWB	4.1	6.5	13.6	15.4	19.7	23.5	24.6	24.2	22.2	18.3	12.5	6.9	(22)	
	5%	DB	3.6	6.2	14.0	20.2	25.6	28.9	30.2	28.7	27.3	22.0	13.8	6.3	(23)	
		MCWB	2.5	4.1	10.4	13.2	18.3	22.1	23.6	23.2	20.6	17.3	10.6	4.7	(24)	
	10%	DB	2.3	3.7	11.0	17.4	22.8	27.4	28.6	27.3	25.1	18.8	11.9	4.0	(25)	
		MCWB	1.5	2.4	7.9	11.4	16.6	21.2	23.1	22.0	19.5	15.3	9.1	2.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	11.3	18.6	19.2	23.6	27.0	28.2	27.7	25.2	22.2	16.2	12.8	(27)	
		MCDB	8.5	12.7	22.0	24.6	28.0	30.4	31.1	30.5	28.3	25.8	18.6	13.7	(28)	
	2%	WB	4.3	7.2	15.0	16.7	21.5	25.1	26.4	25.6	23.5	19.9	13.6	7.5	(29)	
		MCDB	5.6	8.5	17.7	21.4	25.6	28.5	29.6	28.7	27.0	22.9	16.1	8.1	(30)	
	5%	WB	2.7	4.4	11.4	14.5	19.9	23.5	25.1	24.2	22.2	17.9	11.3	5.0	(31)	
		MCDB	3.5	5.9	13.7	19.4	23.9	27.4	28.8	27.1	25.8	20.4	13.4	6.2	(32)	
	10%	WB	1.3	2.6	7.6	12.1	18.1	22.2	23.9	23.1	20.7	15.7	8.8	2.9	(33)	
		MCDB	2.1	3.7	10.0	16.6	21.5	26.1	27.4	26.0	23.5	18.4	11.2	4.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.7	9.7	11.2	11.6	11.2	11.3	10.9	12.1	10.7	8.8	7.5	(35)	
		MCDBR	8.8	10.8	13.6	16.4	14.5	12.7	12.4	12.1	13.9	14.5	12.5	9.1	(36)	
	5% WB	MCWBR	7.7	8.4	9.3	9.4	8.1	6.9	6.6	6.8	7.6	8.8	9.2	7.5	(37)	
		MCWBR	8.1	9.9	12.4	14.2	12.4	11.3	10.9	10.1	11.8	12.1	11.8	8.3	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.279	0.315	0.336	0.382	0.403	0.422	0.419	0.406	0.375	0.337	0.298	0.277	(40)	
		taud	2.385	2.279	2.368	2.278	2.269	2.255	2.263	2.290	2.367	2.470	2.512	2.432	(41)	
	Ebn at Noon	Ebn at Noon	861	875	905	883	871	852	850	849	851	847	838	832	(42)	
		Edh at Noon	77	103	107	127	132	134	132	124	107	85	69	68	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.64	2.68	3.75	4.57	5.38	6.03	6.11	5.28	4.30	2.76	1.78	1.29	(44)	
	RadStd	RadStd	0.23	0.29	0.41	0.40	0.41	0.43	0.29	0.33	0.36	0.32	0.15	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-1.52	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (55 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page